SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve realworld Computer Visio n problems.***		
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Section / Batch:	2023	**Date:**	23-10-26

Code of Conduct

I _ (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

QUESTIONS: 5 Total Marks: 50 Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Histogram Equalization – Step-by-Step Solution

Given pixel intensity values:

10, 10, 20, 20, 20, 30, 30, 40

- Total number of pixels, $N = 8$
- Number of gray levels, $L = 256$
- Histogram equalization formula:

$$ s=\operatorname{round}((L-1)\times CDF) $$

$$ s=\operatorname{round}(255\times CDF) $$

1. Construct the Histogram

Count the frequency of each intensity value.

Intensity (r)	Frequency (f)
10	2
20	3
30	2
40	1
Total	8

2. Calculate the CDF

First, calculate the probability of each intensity:

$$ P(r)=\frac{f}{N} $$

Then, add the probabilities cumulatively to obtain the CDF.

Intensity	Frequency	Probability ($f/8$)	CDF
10	2	0.25	0.25

20	3	0.375	0.625
30	2	0.25	0.875
40	1	0.125	1.000

3. Apply Histogram Equalization

Using the formula:

$$ s=\operatorname{round}(255\times CDF) $$

For intensity 10:

$$ s=\operatorname{round}(255\times0.25) =\operatorname{round}(63.75)=64 $$

For intensity 20:

$$ s=\operatorname{round}(255\times0.625) =\operatorname{round}(159.375)=159 $$

For intensity 30:

$$ s=\operatorname{round}(255\times0.875) =\operatorname{round}(223.125)=223 $$

For intensity 40:

$$ s=\operatorname{round}(255\times1.0)=255 $$

4. New Intensity Values

Original intensity	**CDF**	**New intensity \(s\)**
10	0.25	64
20	0.625	159
30	0.875	223
40	1.000	255

Original pixels:

10, 10, 20, 20, 20, 30, 30, 40

Equalized pixels:

64, 64, 159, 159, 159, 223, 223, 255

5. Compare Contrast Before and After Equalization

Before equalization

Pixel intensities range from 10 to 40.

10 10 20 20 20 30 30 40

After equalization

Pixel intensities spread across a wider range, from 64 to 255.

64 64 159 159 159 223 223 255

Feature	Before	After
Minimum intensity	10	64
Maximum intensity	40	255
Intensity range	30	191
Contrast	Low	Improved

Conclusion

Histogram equalization increases the intensity range from 30 to 191, spreading the pixel values over a wider range. This improves the overall contrast and can make image details easier to distinguish. However, because the image contains only four distinct intensity levels, equalization does not create any new intensity levels or additional image details.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Apply Averaging Filter and Median Filter

Given 3 × 3 image:

$$ I=\begin{bmatrix} 10&20&30\\ 20&100&20\\ 10&20&30 \end{bmatrix} $$

The centre pixel is 100.

1. Apply the 3 × 3 Averaging Filter

The averaging filter calculates the mean of all nine pixel values.

Formula:

$$\text{Average}=\frac{\text{Sum of all pixels}}{9}$$

Substitute the values:

$$=\frac{10+20+30+20+100+20+10+20+30}{9}$$

$$=\frac{260}{9}=28.89$$

Therefore, the new centre pixel value is approximately 29.

New image after averaging:

$$\boxed{ \begin{bmatrix} 10&20&30\\ 20&29&20\\ 10&20&30 \end{bmatrix} }$$

2. Apply the Median Filter

Arrange all nine pixel values in ascending order.

10, 10, 20, 20, 20, 20, 20, 30, 30, 100

Correction: the neighbourhood contains nine values, so the sorted list is:

10, 10, 20, 20, 20, 20, 30, 30, 100

The median is the fifth value.

$$\text{Median}=20$$

Therefore, the new centre pixel value is 20.

New image after median filtering:

$$\boxed{ \begin{bmatrix} 10&20&30\\ 20&20&20\\ 10&20&30 \end{bmatrix} }$$

3. Compare the Results

Feature	Averaging filter	Median filter

Calculation	Mean of pixels	Middle sorted value
New centre value	29 (rounded)	20
Effect on noise	Reduces noise but is affected by extreme values	Removes isolated extreme values effectively
Edge preservation	May blur edges	Generally preserves edges better
Best use	Reducing random noise	Removing impulse (salt-and-pepper) noise

Conclusion

The median filter is better at preserving image details in the presence of impulse noise. The averaging filter is affected by the unusually high centre value of 100, resulting in 29, whereas the median filter replaces it with 20, which is more consistent with its neighbouring pixels.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Apply Laplacian Filter for Image Sharpening

Given grayscale image:

$$ I=\begin{bmatrix} 50&50&50\\ 50&100&50\\ 50&50&50 \end{bmatrix} $$

Laplacian filter mask:

$$ L=\begin{bmatrix} 0&-1&0\\ -1&4&-1\\ 0&-1&0 \end{bmatrix} $$

The original centre pixel intensity is 100.

1. Calculate the Laplacian Response

Multiply each image pixel by its corresponding mask value and add the results.

$$ R=(0\times50)+(-1\times50)+(0\times50) $$

$$ +(-1\times50)+(4\times100)+(-1\times50) $$

$$ +(0\times50)+(-1\times50)+(0\times50) $$

Simplifying:

$$ R=0-50+0-50+400-50+0-50+0 $$

$$ \boxed{R=200} $$

2. Sharpen the Centre Pixel

The question specifies adding the Laplacian response to the original intensity.

Formula:

$$ I_{\text{new}}=I_{\text{original}}+R $$

Substitute the values:

$$ I_{\text{new}}=100+200 $$

$$ \boxed{I_{\text{new}}=300} $$

The resulting centre pixel intensity is 300 before intensity-range clipping. Since a standard 8-bit grayscale image supports values from 0 to 255, the stored value would be clipped to 255.

3. Explain How the Laplacian Filter Enhances Details

- Detects intensity changes: The Laplacian measures differences between the centre pixel and its neighbours.
- Enhances edges: Large intensity differences produce a strong filter response.
- Sharpens fine details: Adding the response increases the contrast of the bright centre pixel relative to its darker surroundings.
- Improves image clarity: Fine structures and boundaries become more noticeable.
- Limitation: Excessive sharpening may amplify noise and cause intensity clipping.

Conclusion

The Laplacian filter response is 200. Adding it to the original centre intensity of 100 produces 300, which is clipped to 255 in an 8-bit image. The filter enhances edges and fine details by emphasizing rapid changes in intensity.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Apply Global Thresholding to a Grayscale Image

Given grayscale image:

$$ I=\begin{bmatrix} 40&60&120\\ 80&150&200\\ 30&100&180 \end{bmatrix} $$

Threshold value: $T=100$

Rule:

- If pixel intensity ≥ 100, assign 1 (foreground).
- If pixel intensity <100, assign 0 (background).

1. Apply Global Thresholding

Check each pixel against the threshold value of 100.

$$ I_b=\begin{bmatrix} 0&0&1\\ 0&1&1\\ 0&1&1 \end{bmatrix} $$

Therefore, the resulting binary image is:

$$ \boxed{\begin{bmatrix} 0&0&1\\ 0&1&1\\ 0&1&1 \end{bmatrix}} $$

2. Count Foreground and Background Pixels

Pixel category	Condition	Number of pixels
Foreground	Intensity ≥100	5
Background	Intensity <100	4

Total	9

Foreground pixels are 120, 150, 200, 100, and 180.

Background pixels are 40, 60, 80, and 30.

3. Explain How Thresholding Works

- Object separation: Global thresholding separates pixels into foreground and background using one fixed threshold.
- Foreground identification: Pixels with intensity values of 100 or above are classified as foreground.
- Background identification: Pixels below 100 are classified as background.
- Simple processing: This method is computationally efficient and easy to implement.

One limitation: Global thresholding may fail when the image has uneven illumination or poor contrast because a single threshold may not correctly separate the object from the background across the entire image.

Conclusion

The binary image contains 5 foreground pixels and 4 background pixels. Global thresholding simplifies image segmentation by converting grayscale values into binary values, but its accuracy depends on selecting an appropriate threshold.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Apply the Sobel Operator for Edge Detection

Given grayscale image:

$$ I=\begin{bmatrix} 10&10&10\\ 10&50&90\\ 10&50&90 \end{bmatrix} $$

Sobel masks:

$$ G_x=\begin{bmatrix} -1&0&1\\ -2&0&2\\ -1&0&1 \end{bmatrix} \qquad G_y=\begin{bmatrix} -1&-2&-1\\ 0&0&0\\ 1&2&1 \end{bmatrix} $$

1. Calculate the Horizontal Gradient (G_x)

Multiply the corresponding pixel values by the G_x mask and add them.

$$ G_x=(-1)(10)+(0)(10)+(1)(10) $$
$$ +(-2)(10)+(0)(50)+(2)(90) $$
$$ +(-1)(10)+(0)(50)+(1)(90) $$

Simplifying:

$$ G_x=-10+10-20+180-10+90 $$
$$ \boxed{G_x=240} $$

2. Calculate the Vertical Gradient (G_y)

Multiply the corresponding pixel values by the G_y mask and add them.

$$ G_y=(-1)(10)+(-2)(10)+(-1)(10) $$
$$ +(0)(10)+(0)(50)+(0)(90) $$
$$ +(1)(10)+(2)(50)+(1)(90) $$

Simplifying:

$$ G_y=-10-20-10+10+100+90 $$
$$ \boxed{G_y=160} $$

3. Calculate the Gradient Magnitude

The gradient magnitude represents the strength of the intensity change.

$$ G=\sqrt{G_x^2+G_y^2} $$

Substitute the calculated values:

$$ G=\sqrt{240^2+160^2} $$
$$ =\sqrt{57600+25600} $$
$$ =\sqrt{83200} $$
$$ \boxed{G\approx288.44} $$

4. Determine the Dominant Edge Direction

Calculate the gradient angle:

$$\theta=\tan^{-1}\left(\frac{G_y}{G_x}\right)$$
$$\theta=\tan^{-1}\left(\frac{160}{240}\right)$$
$$\boxed{\theta\approx33.69^\circ}$$

The gradient points approximately 33.69° from the positive horizontal axis. Since G_x is greater than G_y, the intensity changes more strongly in the horizontal direction, indicating a predominantly vertical edge.

5. Applications of Sobel Edge Detection

- Boundary detection: Identifies object boundaries by detecting strong changes in pixel intensity.
- Image segmentation: Helps separate objects from the background using edge information.
- Feature extraction: Detects shapes, outlines, and structural details in images.
- Computer vision: Supports object recognition, medical image analysis, and vehicle detection.

Conclusion

The Sobel operator produces $G_x=240$, $G_y=160$, and a gradient magnitude of approximately 288.44. The dominant edge is vertical because the horizontal gradient component is larger. Sobel edge detection helps identify object boundaries and supports image segmentation by highlighting regions with significant intensity changes.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature:	Signature:
Date: _____	Date:	Date: _____